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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DSA0308 Natural Language Processing

MODELS AND ALGORITHMS IN NATURAL LANGUAGE PROCESSING (NLP)

Presented by

S Mohammed Abrar (192472185)

Guided by:

Dr. Kumaragurubaran. T



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INTRODUCTION



1. Natural Language Processing (**NLP**) is a branch of **Artificial Intelligence (AI)** that enables computers to understand, interpret, analyze, and generate human language.
2. It combines the fields of **Computer Science (CS)**, **Linguistics**, and **Machine Learning (ML)** to process text and speech efficiently.
3. Models and algorithms are the fundamental building blocks of NLP because they help computers learn language patterns, make predictions, and perform various language-related tasks accurately.



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WHAT ARE NLP MODELS?



An **NLP model** is a computational framework that is trained using large amounts of textual data to understand the structure and meaning of human language. These models learn linguistic patterns and relationships between words to perform specific NLP tasks.

Types of NLP Models

- **Rule-Based Models:** These models use manually designed linguistic rules and dictionaries to process text.
- **Statistical Models:** These models use probability and statistical techniques to predict the next word or classify text.



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WHAT ARE NLP MODELS?



- **Machine Learning (ML) Models:** These models learn patterns from labeled datasets without requiring explicit programming.
- **Deep Learning (DL) Models:** These models use artificial neural networks to capture complex relationships within language.
- **Transformer Models:** These are advanced deep learning models that use the attention mechanism to process entire sentences simultaneously.



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NLP ALGORITHMS



Naïve Bayes (NB)

- Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It is widely used for spam detection, sentiment analysis, and document classification because of its simplicity and efficiency.

Support Vector Machine (SVM)

- Support Vector Machine is a supervised learning algorithm that identifies the optimal decision boundary between different classes. It is commonly used for text categorization and sentiment analysis.

Hidden Markov Model (HMM)

- A Hidden Markov Model is a statistical sequence model that predicts hidden states from observable data. It is extensively used for **Part-of-Speech (POS)** tagging and speech recognition.



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NLP ALGORITHMS



Conditional Random Field (CRF)

- A Conditional Random Field is a probabilistic graphical model used for sequence labeling tasks. It is widely applied in **Named Entity Recognition (NER)** and information extraction.

K-Means Clustering

- K-Means is an unsupervised learning algorithm that groups similar documents into clusters without requiring labeled data.

Term Frequency–Inverse Document Frequency (TF-IDF)

- TF-IDF is a feature extraction technique that measures the importance of a word within a document relative to an entire collection of documents.

Word2Vec

- Word2Vec is a word embedding algorithm that converts words into numerical vectors while preserving their semantic relationships.



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OBJECTIVES



1. The primary objective of Natural Language Processing is to enable computers to understand and communicate using human language effectively.
2. NLP also aims to extract useful information from unstructured text, automate language-related tasks, improve decision-making, generate meaningful responses, and develop intelligent applications capable of interacting naturally with users.



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COMPARISON OF NLP MODELS



Model	Strength	Limitation
Rule-Based	Simple	Limited flexibility
Statistical	Learns probabilities	Requires quality data
Machine Learning	Good accuracy	Needs labeled data
Deep Learning	Learns complex patterns	Computationally expensive
Transformer	State-of-the-art performance	High training cost



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APPLICATIONS



- Natural Language Processing is widely applied across many real-world domains.
- Chatbots and virtual assistants use NLP to understand user queries and generate meaningful responses.
- Machine translation systems convert text from one language to another automatically.
- Sentiment analysis identifies positive, negative, or neutral opinions expressed in customer reviews and social media.
- Spam detection filters unwanted emails using text classification techniques.

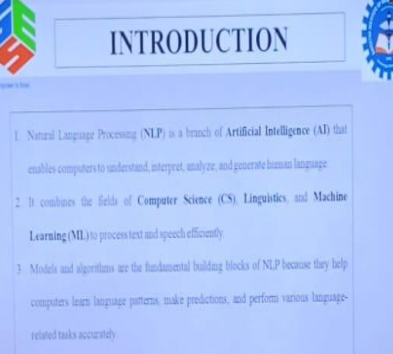


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CONCLUSION



- NLP models and algorithms form the foundation of intelligent language processing.
- Traditional machine learning methods remain useful for many applications, while deep learning and transformer models achieve state-of-the-art performance.
- The continued advancement of NLP is enabling more accurate, efficient, and human-like language understanding across industries.



INTRODUCTION

Dr. Jayaraj P. Sivasankar

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3. Models and algorithms are the fundamental building blocks of NLP because they help computers learn language patterns, make predictions, and perform various language-related tasks accurately.



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Cse & It Department, Mevalurkuppam, Tamil Nadu 602105,
India

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A man in a checkered shirt is standing next to a large screen displaying a presentation slide titled "INTRODUCTION". The slide lists three points about Natural Language Processing (NLP). To the left of the screen is a whiteboard with handwritten text and diagrams. In the foreground, there is a small table with a laptop and other items. The bottom of the image features a Google Maps overlay showing the location of Mevalurkuppam, Tamil Nadu, India, with a red location pin and the Google logo.

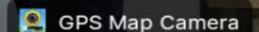


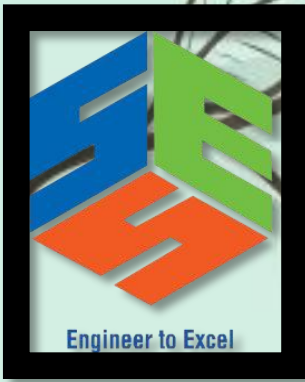
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Thank you